higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

MATHEMATICS N5

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This marking guideline consists of 11 pages.

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QUESTION 1

1.1 $\lim_{x\to 0}\left(\frac{e^x}{e^x-1}-\frac{1}{x}\right)$

$= \lim_{x\to 0}\frac{xe^x-(e^x-1)}{x(e^x-1)}$ ✓ $\frac{0}{0}$

$= \lim_{x\to 0}\frac{xe^x-e^x+e^x}{(e^x-1)+xe^x}$ ✓

$= \lim_{x\to 0}\frac{xe^x}{e^x-1+xe^x}$ $\frac{0}{0}$

$= \lim_{x\to 0}\frac{xe^x+e^x}{e^x+xe^x+e^x}$ ✓

$= \frac{1}{2}$ ✓ (4)

1.2 $f(x)=\frac{x^2-5x+4}{e^{x^2-9}-1}$

$e^{x^2-9}-1=0$

$e^{x^2-9}=e^0$

$x^2-9=0$

$x=\pm 3$ ✓✓ (2)

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QUESTION 2

2.1 $f(x) = \dfrac{5}{7x+2}$

$$f'(x) = \lim_{h\to 0} \frac{\dfrac{5}{7(x+h)+2} - \dfrac{5}{7x+2}}{h} \checkmark$$

$$= \lim_{h\to 0} \frac{\dfrac{5(7x+2) - 5(7x+7h+2)}{(7x+7h+2)(7x+2)}}{h} \checkmark$$

$$= \lim_{h\to 0} \frac{\dfrac{35x+10-35x-35h-10}{(7x+7h+2)(7x+2)}}{h}$$

$$= \lim_{h\to 0} \frac{-35h}{(7x+7h+2)(7x+2)} \times \frac{1}{h} \checkmark$$

$$= \lim_{h\to 0} \frac{-35}{(7x+7h+2)(7x+2)}$$

$$= -\frac{35}{(7x+2)^2} \checkmark \quad (4)$$

2.2 2.2.1 $y = \dfrac{1}{5}\sec^{-4}(4+x^3)$

$$\frac{dy}{dx} = -\frac{4}{5}\sec^{-5}(4+x^3) \checkmark \times \sec(4+x^3)\tan(4+x^3) \checkmark \times 3x^2 \checkmark \quad (3)$$

2.2.1 $y = \dfrac{\sqrt{2x^3+1}}{\sqrt{2x^3-1}}$

$$\frac{dy}{dx} = \frac{\dfrac{1}{2\sqrt{2x^3+1}} \checkmark \times 6x^2 \checkmark \times \sqrt{2x^3-1} - \dfrac{1}{2\sqrt{2x^3-1}} \checkmark \times 6x^2 \checkmark \times \sqrt{2x^3+1}}{\left(\sqrt{2x^3-1}\right)^2} \quad (4)$$

2.3 $y = \dfrac{(3x+5)^3}{\sqrt{x^4-1}}$

$\ln y = 3\ln(3x+5) - \dfrac{1}{2}\ln(x^4-1)$

$\dfrac{1}{y}\dfrac{dy}{dx}\checkmark = 3 \times \dfrac{1}{3x+5} \times 3\checkmark - \dfrac{1}{2} \times \dfrac{1}{x^4-1} \times 4x^3\checkmark$

$\dfrac{dy}{dx} = y\left(\dfrac{9}{3x+5} - \dfrac{2x^3}{x^4-1}\right)\checkmark$

$\dfrac{dy}{dx} = \dfrac{(3x+5)^3}{\sqrt{x^4-1}}\left(\dfrac{9}{3x+5} - \dfrac{2x^3}{x^4-1}\right)$ (4)

2.4 2.4.1 $(y^2+x^2)^3 = 8x^2y^2$

$3(y^2+x^2)^2 \times \left(2x + 2y\dfrac{dy}{dx}\right)\checkmark = 16xy^2 + 16x^2\dfrac{dy}{dx}\checkmark$

$6x(y^2+x^2)^2 + 6y(y^2+x^2)^2\dfrac{dy}{dx} = 16xy^2 + 16x^2y\dfrac{dy}{dx}$

$[6y(y^2+x^2)^2 - 16x^2y]\dfrac{dy}{dx} = 16xy^2 - 6x(y^2+x^2)^2\checkmark$

$\dfrac{dy}{dx} = \dfrac{16xy^2 - 6x(y^2+x^2)^2}{6y(y^2+x^2)^2 - 16x^2y}\checkmark$ (4)

2.4.2 $m = \dfrac{dy}{dx} = \dfrac{16(-1)(1)^2 - 6(-1)((1)^2+(-1)^2)^2}{6(1)((1)^2+(-1)^2)^2 - 16(-1)^2(1)} = 1\checkmark$

$y = x + c$

$-1 = 1 + c$

$c = -2\checkmark$

$y = x - 2\checkmark$ (3)

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QUESTION 3

3.1 3.1.1 $f(x) = x^3 - 2x + 3$

$f'(x) = 3x^2 - 2 = 0$

$x = \pm\sqrt{\frac{2}{3}}$

$x = 0{,}817 \quad or \quad x = -0{,}817$

$y = 1{,}911$ ✓ $y = 4{,}089$

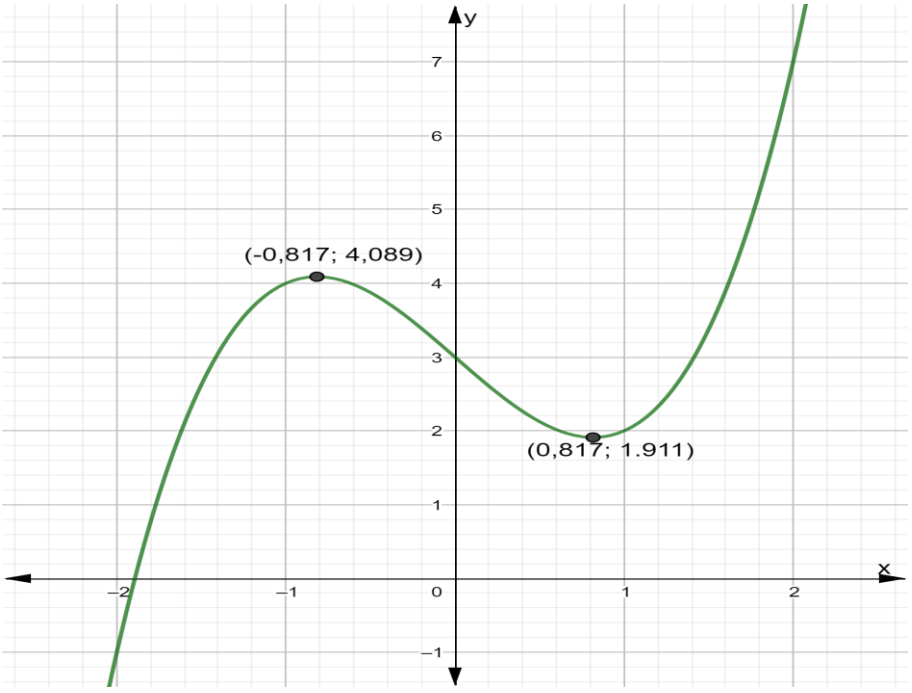
Thus, the coordinates of the turning points are $(0{,}817\ ;\ 1{,}911)$✓ and $(-0{,}817; 4{,}089)$.✓ (3)

3.1.2

x	-2	-1	0	1	2
y	-1	4	3	2	7

✓✓= 2 Marks for the any 2 correct answers. (2)

3.1.3



✓ =1 Mark for the shape.
✓=1 Mark for the indication of the turnings point on the graph. (2)

3.1.4 Let $x_0 = -1{,}9$

$f(-1{,}9) = -0{,}059$

$f'(-1{,}9) = 8{,}830$✓

$x_1 = -1{,}9 - \frac{-0{,}059}{8{,}830}$✓

$= -1{,}893$✓ (3)

3.2 $xy = 256$✓

$$y = \frac{256}{x}$$

$S = 4x + y$✓

$$= 4x + \frac{256}{x}$$

$$= 4x + 256x^{-1}$$

$S' = 4 - 256x^{-2} = 0$✓
$4x^2 - 256 = 0$
$x^2 = 64$;
$x = \pm 8$

$\therefore x = 8$✓ $y = 32$✓ (5)

3.3 $V = s^3$

$$s = \sqrt[3]{V}$$

$A = 6s$

$A = 6\sqrt[3]{V}$✓

$$\frac{dA}{dt} = 6 \times \frac{2}{3} \times \frac{1}{\sqrt[3]{V}} \frac{dV}{dt}$$✓

$$= \frac{4}{\sqrt[3]{V}} \frac{dV}{dt}$$✓

$$\frac{dA}{dt} = \frac{4}{\sqrt[3]{15^3}}(-9)$$✓

$= -2{,}4 cm^2/s$✓ (5)
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QUESTION 4

4.1 4.1.1 $\int \frac{\tan^4\sqrt{x}.\sec^2\sqrt{x}}{\sqrt{x}}dx$

$let\ u = \tan\sqrt{x}$
$2du = \frac{\sec^2\sqrt{x}}{\sqrt{x}}dx$✓

$= 2\int u^4 du$ ✓

$= \frac{2}{5}u^5 + C$

$= \frac{2}{5}(\tan\sqrt{x})^5$✓ $+ C$ (3)

4.1.2 $\int \cos 2x \cos(-6x)\,dx$

$= \int \frac{1}{2}[\cos(2x-6x) + \cos(2x+6x)]\,dx$

$= \frac{1}{2}\int (\cos -4x + \cos 8x)\,dx$✓

$= \frac{\sin 3x}{8}$✓ $+ \frac{\sin 8x}{16}$✓C (3)

4.1.3 $\int \sin^2\left(\frac{15}{4}x\right)dx$

$= \frac{1}{2}\int \left[1 - \cos\left(\frac{15}{2}x\right)\right]dx$ ✓

$= \frac{1}{2}x$✓ $- \frac{1}{15}\sin\left(\frac{15}{2}x\right)$✓ $+ C$ (3)

4.1.4 $\int \frac{x^3 + 9x^2 + 24x + 16}{x-1}dx$

$= \int \left(x^2 + 10x + 34 + \frac{50}{x-1}\right)dx$ ✓

$= \frac{1}{3}x^3$✓ $+ 5x^2$✓ $+ 34x$✓ $+ 50\ln(x-1)$✓ $+ C$ (5)

4.1.5 $\int \cos x \ln(\tan x)\, dx$

$let\ u = \ln(\tan x) \qquad dv = \cos x\, dx$

$du = \frac{1}{\tan x} \times \sec^2 x\, dx \qquad v = \sin x$ ✓

$= \frac{1}{\sin x}.\frac{1}{\cos x}\, dx$

$\int \cos x \ln(\tan x)\, dx = \sin x \ln(\tan x) - \int \sin x \times \frac{1}{\sin x}.\frac{1}{\cos x}\, dx$ ✓

$= \sin x \ln(\tan x) - \int \sec x\, dx$ ✓

$= \sin x \ln(\tan x) - \ln(\tan x + \sec x)$ ✓ $+ C$ (4)

4.2 $\int \frac{x^2 + 2x + 1}{(2x-3)^3}\, dx$

$\frac{x^2 + 2x + 1}{(2x-3)^3} = \frac{A}{(2x-3)^3} + \frac{B}{(2x-3)^2} + \frac{C}{2x-3}$

$x^2 + 2x + 1 = A + B(2x-3) + C(2x-3)^2$ ✓

$x^2 + 2x + 1 = 4Cx^2 + (2B - 12C)x + (A - 3B + 9C)$

$4C = 1; \qquad C = \frac{1}{4}$ ✓

$2B - 12C = 2; \qquad B = \frac{5}{2}$ ✓

$A - 3B + 9C = 1, \qquad A = \frac{25}{4}$ ✓

$\int \left(\frac{\frac{25}{4}}{(2x-3)^3} + \frac{\frac{5}{2}}{(2x-3)^2} + \frac{\frac{1}{4}}{2x-3} \right) dx$

$= -\frac{25}{16}\frac{1}{(2x-3)^2}$ ✓ $- \frac{5}{4}\frac{1}{(2x-3)}$ ✓ $+ \frac{1}{8}\ln(2x-3)$ ✓ $+ C$ (7)

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QUESTION 5

5.1 $\int_1^5 x^4 \ln x \, dx$

$let\ u = \ln x \qquad dv = x^4\,dx$
$du = \frac{1}{x}dx \qquad v = \frac{1}{5}x^5$ ✓

$$\int_1^5 x^4 \ln x \, dx = \frac{1}{5}x^5 \ln x - \int_1^5 \frac{1}{5}x^5 . \frac{1}{x} dx \ \checkmark$$

$$= \frac{1}{5}x^5 \ln x - \frac{1}{5}\int_1^5 x^4\,dx \ \checkmark$$

$$= \left[\frac{1}{5}x^5 \ln x - \frac{1}{25}x^5\right]_0^5 \checkmark$$

$$= \left[\frac{1}{5}(5)^5 \ln 5 - \frac{1}{25}(5)^5\right] - \left[\frac{1}{5}(1)^5 \ln 1 - \frac{1}{25}(1)^5\right] = 880{,}939 \checkmark$$ (5)

5.2 $\mathcal{L}\left(\frac{t}{7}\right) = \frac{1}{7s^2}$ (1)

5.3 5.3.1

$$A = \int_a^b (y_1 - y_2)\,dx$$

$$= \int_0^6 \left[\left(9 - \frac{x^2}{4}\right) - \left(9 - \frac{3}{2}x\right)\right] dx$$

$$= \int_0^6 \left(-\frac{x^2}{4} + \frac{3}{2}x\right) dx \checkmark$$

$$= \left[-\frac{1}{12}x^3 + \frac{3}{4}x^2\right]_0^6 \checkmark$$

$$= 9\ units^2 \quad \checkmark$$ (3)

5.3.2

$$V = \pi \int_a^b (y_1{}^2 - y_2{}^2)\, dx$$

$$= \pi \int_0^6 \left[\left(9 - \frac{x^2}{4}\right)^2 - \left(9 - \frac{3}{2}x\right)^2\right] dx$$

$$\pi \int_0^6 \left(81 - \frac{9}{2}x^2 + \frac{1}{16}x^4 - 81 - \frac{9}{4}x^2 + 27x\right) dx \checkmark\checkmark$$

$$= \pi \int_0^6 \left(\frac{1}{16}x^4 - \frac{27}{4}x^2 + 27x\right) dx \checkmark$$

$$= \pi \left[\frac{1}{80}x^5 - \frac{9}{4}x^3 + \frac{27}{2}x^2\right]_0^6 \checkmark$$

$$= \frac{485\pi}{2} units^3 \; or \; 305{,}363\, units^3 \checkmark$$ (5)

5.4 $I_y = r^2\, dA$

$dA = 8dx$ ✓

$I_y = x^2.8\, dx$✓

$$= 8\int_0^{12} x^2\, dx\checkmark$$

$$= 8\left[\frac{x^3}{3}\right]_0^{12} \checkmark$$

$$= 4608\, units^4\checkmark$$ (5)

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QUESTION 6

6.1

$$x\frac{dy}{dx} = x + 1 + xy + y$$

$$x\frac{dy}{dx} = (x+1)(y+1)$$

$$\frac{dy}{y+1} = \frac{x+1}{x}\, dx\checkmark$$

$$\frac{dy}{y+1} = \left(1 + \frac{1}{x}\right) dx$$

$\ln(y+1)\checkmark = x + \ln x + C \checkmark$ (3)

6.2 $\frac{1}{(2-5x)^2}\frac{d^2y}{dx^2} = -3$, if $\frac{dy}{dx}_{|\ x=1} = 0$ and $y(1) = -\frac{1}{4}$

$$\frac{d^2y}{dx^2} = -75x^2 + 60x - 12$$

$$\frac{dy}{dx} = -\frac{75}{3}x^3 + 30x^2 - 12x + A\checkmark$$

$$0 = -\frac{75}{3}(1)^3 + 30(1)^2 - 12(1) + A$$

$$A = 7\checkmark$$

$$\frac{dy}{dx} = -\frac{75}{3}x^3 + 30x^2 - 12x + 7$$

$$y = -\frac{75}{12}x^4 + 10x^3 - 6x^2 + 7x + B\checkmark$$

$$-\frac{1}{4} = -\frac{75}{12}(1)^4 + 10(1)^3 - 6(1)^2 + 7(1) + B$$

$$B = -5\checkmark$$

$$y = -\frac{75}{12}x^4 + 10x^3 - 6x^2 + 7x - 5\checkmark$$

(5)
[8]

TOTAL: 100

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